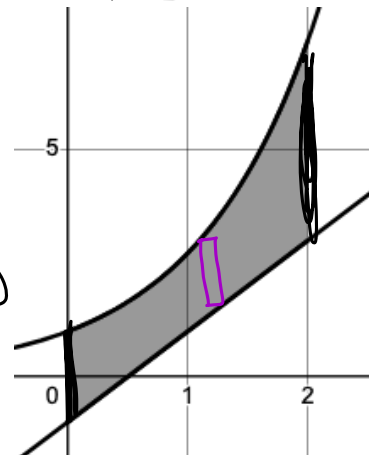


Each problem is worth 10 points. You must show your work and clearly indicate your answers to get credit.

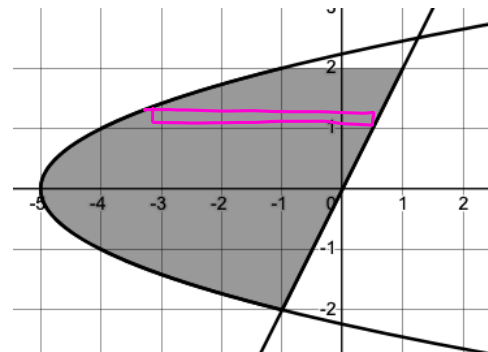
1. Find the shaded area. Give an exact answer. The boundaries are the y -axis, $x = 2$, $y = 2x - 1$ and $y = e^x$.

$$\begin{aligned} & \int \text{TOP-Bottom} \\ & \int_0^2 e^x - (2x - 1) dx \\ & = \left[e^x - x^2 + x \right]_0^2 = (e^2 - 4 + 2) - (1 - 0 - 0) \\ & = e^2 - 2 - 1 = \boxed{e^2 - 3} \end{aligned}$$



2. Find the area of the shaded region. The boundaries are the line $x = \frac{1}{2}y$, the curve $x = y^2 - 5$, and the horizontal line $y = 2$.

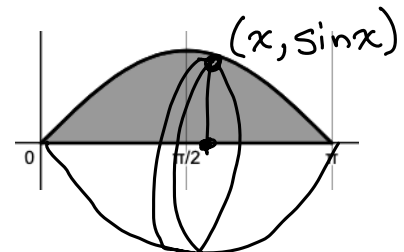
$$\begin{aligned} & \int_{-2}^2 \text{Right-Left} = \int_{-2}^2 \left(\frac{1}{2}y - y^2 + 5 \right) dy \\ & = \left[\frac{1}{4}y^2 - \frac{1}{3}y^3 + 5y \right]_{-2}^2 \\ & = \left(\frac{1}{4}(4) - \frac{1}{3}(8) + 10 \right) - \left(\frac{1}{4}(4) + \frac{8}{3} - 10 \right) \\ & = \boxed{\frac{44}{3}} \end{aligned}$$



3. The Area shown here is rotated around the x -axis. Set up an integral for the volume of the resulting solid. **Do not evaluate the integral.**

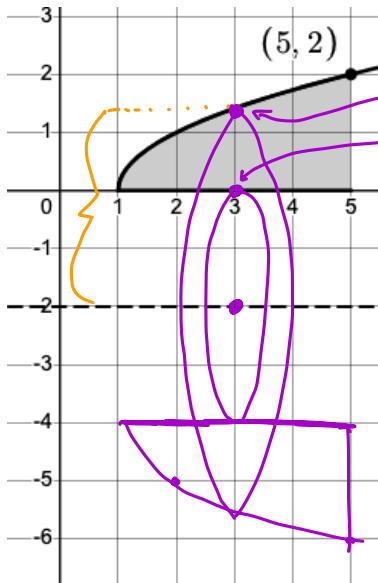
$$V_{Disk} = \pi (\text{rad})^2 = \pi \sin^2 x$$

$$V = \int_0^{\pi} \pi \sin^2 x dx$$



Problems 4-6 use the region bounded by $y = \sqrt{x-1}$, the x -axis, and $x = 5$.

4. The region is rotated around the line $y = -2$. Set up an integral for the volume. **Do not evaluate the integral.**

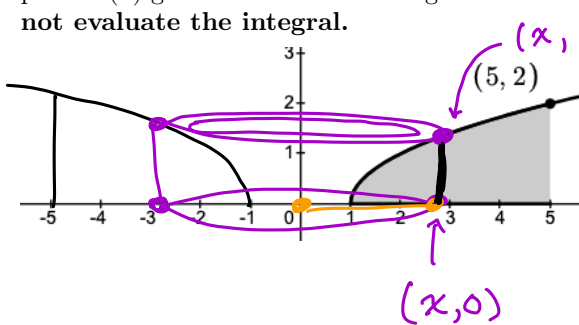


$$V = \pi \int R^2 - r^2 dx$$

$$R = 2 + \sqrt{x-1}$$

$$V = \pi \int_1^5 (2 + \sqrt{x-1})^2 - 4 dx$$

5. The region is rotated around the y -axis. (a) Draw in a SHELL and label the coordinates of the important points. (b) give formulas for the height and radius of the shell. (c) Then set up an integral for its volume. **Do not evaluate the integral.**



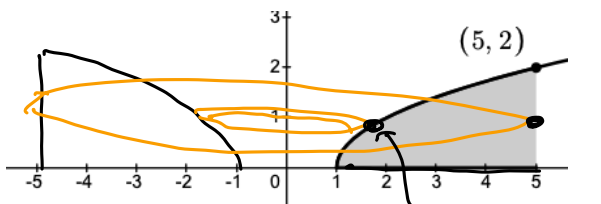
$$V_{\text{shell}} = 2\pi (\text{rad})(\text{height})(\text{thick})$$

$$2\pi \int_1^5 x \sqrt{x-1} dx$$

$$\text{rad} = x$$

$$\text{height} = \sqrt{x-1}$$

6. The region is again rotated around the y -axis. (a) Draw in a WASHER and label the coordinates of the important points. (b) give formulas for the large and small radii. (c) Then set up an integral for its volume. **Do not evaluate the integral.**



$$\text{thick} = dy$$

$$V = \pi \int R^2 - r^2 dy$$

$$= \pi \int_0^2 5^2 - (y^2 + 1)^2 dy$$

$$R = 5$$

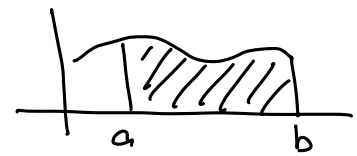
$$r = y^2 + 1$$

$$y = \sqrt{x-1}$$

$$y^2 = x-1$$

$$y^2 + 1 = x$$

$$\text{Avg} = \frac{1}{b-a} \int_a^b f(x) dx$$



7. Find the average value of the function $f(x) = x^2 - 4x$ on the interval $[3, 6]$

$$\frac{1}{6-3} \int_3^6 x^2 - 4x dx$$

$$A = \text{width} \cdot (\text{Avg height})$$

$$\int_a^b f(x) dx = b-a$$

$$\text{Avg height} = \frac{1}{b-a} \cdot \text{Area}$$

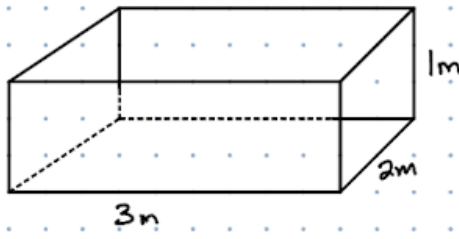
~~8.~~ A spring has a constant of $k = 10$ newtons/meter. Its natural length is 20cm. Use Hooke's law $F(x) = kx$ to answer the following:

(a) How much force is required to hold the spring stretched to a length of 35cm?

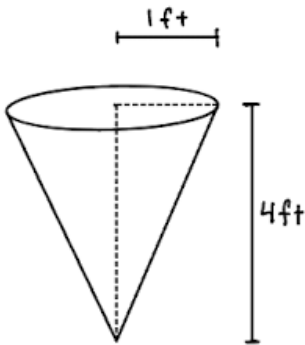
(b) How much work is required to stretch the spring from its natural length to a length of 50 cm? (There's way more space here than you should need)

Note: The weight density of water is 62.5 pounds per square foot, or 9800 Newtons per cubic meter.

9. The aquarium below is full of water. How much work would it take to pump all of the water out of the top of the tank?



-
10. A conical tank is 4 feet tall and has a radius of 1 foot at the top. How much work is required to pump the water out of the top of the tank? Round to the nearest pound-foot.



Show your work and circle or box your answers.

1. Match each integral with the method that should be used to evaluate it. Use each letter exactly once. Write the letter in the blank.

(2 pts each)

C $\int \frac{x^3}{\sqrt{9-x^4}} dx$ $\begin{matrix} u=9-x^4 \\ du=-4x^3 dx \\ -\frac{1}{4} du = x^3 dx \end{matrix}$

(A) regular integration by parts (IBP)

B $\int \frac{x^2}{\sqrt{9-x^2}} dx$ $\begin{matrix} u=9-x^2 \\ du=-2x dx \end{matrix}$

(B) trig substitution

A $\int \sin x \ln x dx$

(C) u -substitution

D $\int x^2 \sin(5x) dx$

(D) Tabular IBP

F $\int \sin(x)e^{2x} dx$

(E) partial fractions

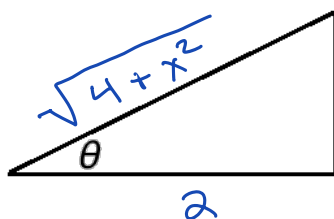
E $\int \frac{2x+4}{x^2-1} dx$

(F) IBP twice & solve

2. Factor the denominator and write out the partial fraction expansion of the function below. Don't solve for the coefficients. (8 pts)

$$\frac{x^2 + x - 5}{(x+5)(x^4 + 9x^2)}$$

3. Suppose that you used the trig substitution $x = 2 \tan \theta$ to rewrite an integral, and after integrating with respect to θ , you got $\theta + \ln |\sec \theta| + C$ for your answer. Put this answer back in terms of x . Be sure to label the sides of the triangle. (8 pts)



$$x = \tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{x}{2}$$

$$\sec \theta = \frac{\text{hyp}}{\text{adj}} = \frac{\sqrt{4+x^2}}{2}$$

$$\theta = \arctan\left(\frac{x}{2}\right)$$

$$\arctan\left(\frac{x}{2}\right) + \ln \left| \frac{\sqrt{4+x^2}}{2} \right| + C$$